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Towards Personalized Fitness-to-drive Assessment using Driver Digital Twins

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Abstract

Assessing whether a driver is fit to drive remains a critical yet largely unsolved problem in road safety and intelligent transportation. Existing monitoring systems rely on generic models that fail to account for individual differences in age, experience, and emotional state, limiting both their accuracy and practical utility. In this paper, we present a personalized Fitness-to-Drive (FtD) index computed through a Driver Digital Twin (DrDT), i.e., a cyber-physical system in which the driver constitutes the physical entity, continuously synchronized with a virtual counterpart through an IoT sensor pipeline. The DrDT is formalized as a domain-specific instantiation of the Human Digital Twin framework, integrating real-time physiological and behavioral signals, including heart rate, facial emotion recognition, and visual distraction, with a structured driver profile to dynamically adapt the FtD assessment to each individual. The system is implemented as an event-driven microservice architecture leveraging MQTT for sensor ingestion, RabbitMQ for service coordination, and a hybrid MongoDB/Redis storage layer for real-time and historical data access. We validate our approach through controlled experiments with 20 drivers using the CARLA simulator, demonstrating that personalization reduces mean absolute error by over 60% (from 0.11 to 0.04) and RMSE by over 55% (from 0.12 to 0.05) compared to a generic baseline. Gains are largest for drivers whose profiles deviate most from the population mean, confirming the value of individualization. We discuss implications for the design of adaptive cyber-physical driver monitoring systems, along with open challenges in IoT sensor integration, privacy, security, and real-world deployment.

Keywords: Digital twin, Driver monitoring, Personalized safety assessment, Driver digital twin, Edge computing

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1. Introduction

Assessing driver fitness in real time remains a critical yet unsolved challenge at the intersection of road safety and cyber-physical systems. Road traffic accidents represent one of the leading causes of death and injury worldwide, with human factors including, e.g., fatigue, distraction, and emotional state, accounting for the vast majority of incidents [1,2]. Despite decades of research, existing monitoring systems still rely on generic models that ignore individual differences in age, experience, and psychological state, leaving drivers without timely, personalized feedback on their actual risk level.

The emergence of IoT-enabled vehicles and cyber-physical systems opens a new path forward. Modern vehicles are increasingly instrumented with sensors that continuously stream physiological, behavioral, and contextual data, creating the infrastructure for real-time, individualized driver assessment. Digital Twins (DTs), virtual replicas of physical entities continuously synchronized with real-world sensor data through bidirectional communication channels [3,4], are a natural architectural fit for this setting: the driver constitutes the physical entity, the IoT pipeline provides the communication layer, and the twin maintains a continuously updated cyber representation of the driver's state. Applied to individuals, they enable real-time monitoring of physiological parameters, behavioral patterns, and environmental interactions [5,6], providing the foundation for genuinely personalized safety assessment. Recent work has begun exploring this direction, from in-vehicle data recorders that generate aggregate risk scores [1] to DT frameworks for autonomous vehicle development [7], yet no existing approach delivers a fully personalized, real-time Fitness-to-Drive assessment grounded in a rigorous Human Digital Twin (HDT) model.

In this paper, we present a Driver Digital Twin (DrDT) framework for computing a personalized Fitness-to-Drive (FtD) index. The DrDT is a cyber-physical system where an IoT sensor pipeline, spanning wearable heart rate monitoring, in-cabin camera-based emotion and distraction detection, and vehicle telemetry, continuously synchronizes a virtual driver model with the physical driver's real-time state. Built as a domain-specific instantiation of the Human Digital Twin [8,9], the DrDT integrates these real-time signals with a structured driver profile capturing age, experience, and annual mileage, and implements the full pipeline as an event-driven microservice architecture using MQTT, RabbitMQ, and a hybrid MongoDB/Redis storage layer. Specifically, this paper makes the following contributions:

- A unified definition of Driver Digital Twin (DrDT) as a domain-specific instantiation of the HDT framework, formalizing the driver as a cyber-physical entity whose transient states and enduring characteristics are continuously represented and updated in a virtual counterpart.
- A personalized Fitness-to-Drive (FtD) index that dynamically adapts to individual driver characteristics, e.g., age, experience, emotional arousal, and distraction, rather than relying on population-level assumptions.
- An IoT microservice implementation of the DrDT and FtD pipeline, leveraging MQTT for high-frequency sensor ingestion, RabbitMQ for reliable service-to-service coordination, and a hybrid MongoDB/Redis storage layer for real-time and historical data access.
- An experimental validation using the CARLA simulator with 20 drivers, demonstrating over 60% reduction in mean absolute error compared to a generic baseline, with personalization yielding the greatest gains for drivers furthest from the population mean.

The paper is organized as follows. Section 2 reviews related work on FtD assessment and digital twin technologies. Section 3 introduces our DrDT definition, design and implementation. Section 4 presents experimental results. Section 5 discusses findings and limitations and Section 6 concludes.

2. Background and State of the Art

This section reviews the two pillars underlying our work: Fitness-to-Drive (FtD) assessment and Digital Twin (DT) technologies applied to individuals. For each, we identify the key findings and the gaps that motivate our contribution.

2.1. Fitness-to-Drive

The FtD refers to a driver's ability to operate a vehicle safely given their physical, cognitive, and emotional state. Most existing assessments occur off-road, at license issuance or renewal, using regression or classification models trained on predetermined populations, typically individuals with brain injuries, Alzheimer's, or motor disabilities [10–12]. While valuable, these approaches are inherently static and population-level, offering no real-time, personalized assessment during active driving.

On-road approaches have attempted to close this gap. Dronseyko et al. [13] proposed a driving hazard coefficient based on speed and lateral distance from safe reference values. Toledo et al. [1] demonstrated that in-vehicle data recorders (IVDRs)[1] can significantly reduce accident rates by providing drivers with risk feedback derived from acceleration, speed, and GPS data. Ji et al. [2] addressed fatigue specifically, using facial feature analysis and heart rate to predict drowsiness in real

[1] IVDR stands for "In-Vehicle Data Recorder", an electronic device installed within a vehicle that records and stores data related to driving and driver behavior.

time. Yi-Cheng et al. [14] proposed a driving assistance system integrating vehicle dynamics from On-Board Diagnostics (OBD-II) and real-time traffic conditions to reduce accident risk and improve safety through driver feedback. More recently, our own prior work [15] introduced a real-time FtD index integrating visual distraction, cognitive distraction, and vehicle speed, considering the six primary emotions identified by Ekman [16] plus a neutral state. A further refinement [17] weighted these factors by their relative impact on driving performance and conditioned emotional penalties on arousal level.

Despite this progress, none of these approaches personalizes the FtD assessment to the individual driver. Generic models apply the same weights and thresholds to all drivers, ignoring systematic differences in age, experience, and baseline physiological response. This is the gap our work directly addresses.

2.2. Factors Affecting Driving Fitness

Driving fitness is shaped by both internal and external factors [18]. Internal factors, e.g., age, gender, experience, personality, and cognitive and physical state, are particularly relevant to personalization, as they vary significantly across individuals and are directly measurable through a driver profile. External factors such as weather, traffic conditions, and vehicle design also play a role but fall outside the scope of the present work.

Age and experience are the most consequential internal factors. Young drivers and elderly drivers are disproportionately involved in accidents: young drivers due to limited hazard perception, elderly drivers due to cognitive and sensory decline [19,20]. Summala [21] further shows that experience builds skills and strategies that reduce anxiety and improve decision-making. Upahita et al. [22] demonstrate that hazard mitigation skills peak at three to five years of experience and deteriorate rapidly with driving inactivity, particularly in the first two to three years after license acquisition. This pattern is corroborated by broader analyses of risky behavior across age groups and cities [23,24]. Annual mileage, as a proxy for active experience, is also a significant predictor of accident involvement [25].

These findings directly inform our Driver Profile design: age, years of experience, and annual mileage are the primary static factors used to personalize the FtD index, with the greatest accuracy gains expected for drivers whose profiles deviate most from the population mean.

2.3. Digital Twins and Human Digital Twins

A DT is a virtual replica of a physical entity, continuously updated with real-world data to simulate, analyze, and predict its behavior [26–28]. First conceived by Grieves in 2003 [29], DTs require three essential components: a physical reality, a virtual representation, and interconnections enabling data exchange between the two [3]. What distinguishes a DT from a conventional simulator is bi-directional communication: data flows not only from the physical entity to the virtual model, but also back, enabling virtual insights to influence the physical system [30]. DTs are now applied across manufacturing, aerospace, healthcare, and automotive domains [31], benefiting from advances in sensors, cloud computing, and artificial intelligence to maintain accurate, real-time representations [4].

The Human Digital Twin (HDT) extends this concept to individuals, representing a person's physiological and psychological state in a virtual space [8,9]. HDTs enable real-time monitoring of physiological parameters and interactions with the environment [5,6], and have been explored in healthcare for personalized therapy [32] and in transport — notably in a US Air Force project creating a DT of a fighter pilot to support personalized training and injury prevention [7]. In the automotive domain, Hu et al. [33] introduced the Driver Digital Twin (DrDT) concept, integrating data-driven and physics-based models to simulate driving behavior and predict driver intentions.

A key enabler of HDTs is the user profile, a structured, dynamic representation of personal characteristics including demographics, preferences, and behavioral patterns [34–38]. Profiles can be initialized using archetypes, which encode typical characteristics of a user group to bootstrap the system before sufficient individual data is available, a form of knowledge-based initialization analogous to pre-trained models in machine learning. The system then refines these defaults as real observations accumulate [39–41]. Profile updates can be driven by data mining and ontology-based mechanisms that learn implicit user preferences over time [36], progressively improving personalization accuracy.

However, existing DrDT work focuses on behavioral simulation and intention prediction rather than real-time safety assessment. No prior work has formalized a DrDT definition grounded in HDT principles and used it to compute a personalized, continuously updated FtD index. This is precisely the contribution of the present paper.

3. Proposed Model and Implementation

We pursue two tightly coupled objectives. First, we formalize a general definition of Driver Digital Twin (DrDT) as a domain-specific instantiation of the Human Digital Twin (HDT) framework, grounding it explicitly in a cyber-physical architecture in which the driver is the physical entity, an IoT sensor pipeline is the communication layer, and the twin is the continuously synchronized cyber representation. Second, we exploit this definition to compute a personalized Fitness-to-Drive (FtD) index that dynamically adapts to each driver's profile, physiological state, and behavioral patterns. As we demonstrate in Section 4, this integration yields substantial accuracy gains over generic, population-level approaches.

3.1. Driver Digital Twin: Definition

We extend the HDT concept to the automotive domain by introducing the DrDT. Unlike the general HDT, designed to represent a human across arbitrary contexts, the DrDT specifically captures both the transient states (e.g., current emotional arousal, distraction level) and the enduring characteristics (e.g., age, experience) of a driver during active vehicle operation.

Definition 1 (Driver Digital Twin)**.** *A Driver Digital Twin is a tuple (C_S, E_S, D_P, D_S, C), where:*

- C_S ***(Cognitive State)*** *represents the driver's visual and cognitive distraction levels, inferred via in-vehicle sensors and a Driver Monitoring System (DMS);*
- E_S ***(Emotional State)*** *captures the driver's affective condition in terms of arousal and valence, also inferred by the DMS;*
- D_P ***(Driver Profile)*** *encodes static and semi-static personal attributes, modeled as a mapping from characteristic identifiers to their current values. This profile both conditions the interpretation of real-time signals and drives the personalization of system responses;*
- D_S ***(Driving Style)*** *denotes the driver's behavioral style class (e.g., reckless, anxious, patient and careful [42]), enabling the system to tailor feedback and interventions accordingly;*
- C ***(Confidence)*** *quantifies the driver's trust in the vehicle's autonomous systems as a continuous value in $[0, 1]$, used to calibrate alert thresholds and adaptive responses.*

Definition 1 is intentionally application-agnostic: it abstracts away sensor-level implementation details so that the DrDT can be instantiated across diverse automotive contexts. Viewed through a cyber-physical systems lens, C_S and E_S represent the high-frequency sensing layer of the physical-to-cyber data flow, D_P encodes the persistent cyber state of the individual driver, and D_S and C govern the cyber-to-physical feedback loop through alert calibration and adaptive interventions. The implementation presented in this work realizes a subset of this definition, i.e., driving style D_S is not modeled, as it is not required for FtD computation, but the full tuple provides a foundation for richer future instantiations. All privacy requirements inherent to HDT apply unchanged.

3.2. Driver Digital Twin: Design

Figure 1 illustrates our DrDT implementation. The model integrates three information streams, cognitive state, emotional state, and driver profile, into a unified driver state representation, which feeds directly into the FtD computation.

Cognitive and emotional states are inherently high-frequency signals. Cognitive state is inferred from heart rate variability measured via smartwatch, while emotional state and visual distraction are

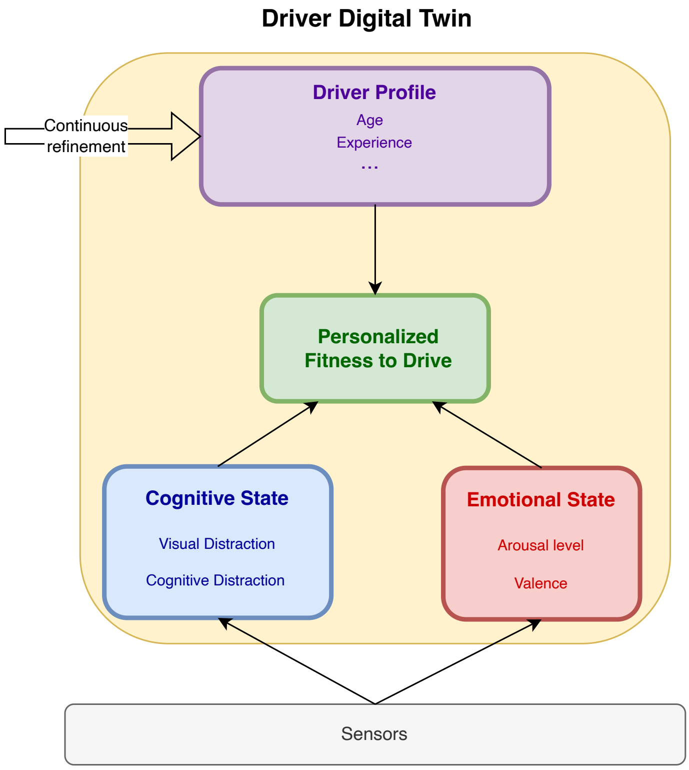

Figure 1. Proposed DrDT model.

detected through continuous facial analysis from an in-cabin camera. Both streams are updated in real time and published over MQTT to the processing pipeline.

Driver Profile is modeled following the Virtual User framework of [40,41], adapted to the driving context. Since the User Context is fixed (i.e., always driving), we focus exclusively on the User Profile, here referred to as Driver Profile. It is structured as a set of key-value mappings across two update tiers:

- **General data**, e.g., name, date of birth, license acquisition date, and the driver's unique DT identifier. These attributes have the lowest update frequency and are typically set at registration.
- **Characteristics**, e.g., annual kilometers driven, visual acuity, and similar behavioral or physiological attributes. These are updated at medium frequency: mileage annually, visual acuity over months.

A key design choice is the *dynamic* management of profile attributes: the DrDT continuously refines its internal representation in response to observed behavioral patterns and environmental context. This evolution is what enables the FtD index to remain genuinely personalized over time rather than relying on a static snapshot of the driver.

3.3. Personalized Fitness-to-Drive Index

Building on prior work [15,17], we introduce a personalized FtD index that leverages the full DrDT state. The core innovation is a Personal Factor (PF) derived from the Driver Profile, which modulates the contribution of distraction and emotion relative to the individual driver's characteristics. The index is defined as:

$$FtD = 1 - f(V) \cdot (w_D \cdot DF + w_E \cdot EF + w_P \cdot PF) \tag{1}$$

where $f(V) = 1 + k \cdot V$ or $f(V) = e^{k \cdot V}$ is a speed-scaling function with factor k, and the weights w_D, w_E, w_P satisfy $w_D + w_E + w_P = 1$. A value of $FtD = 1$ indicates full fitness; lower values reflect increasing impairment. The three components are detailed below.

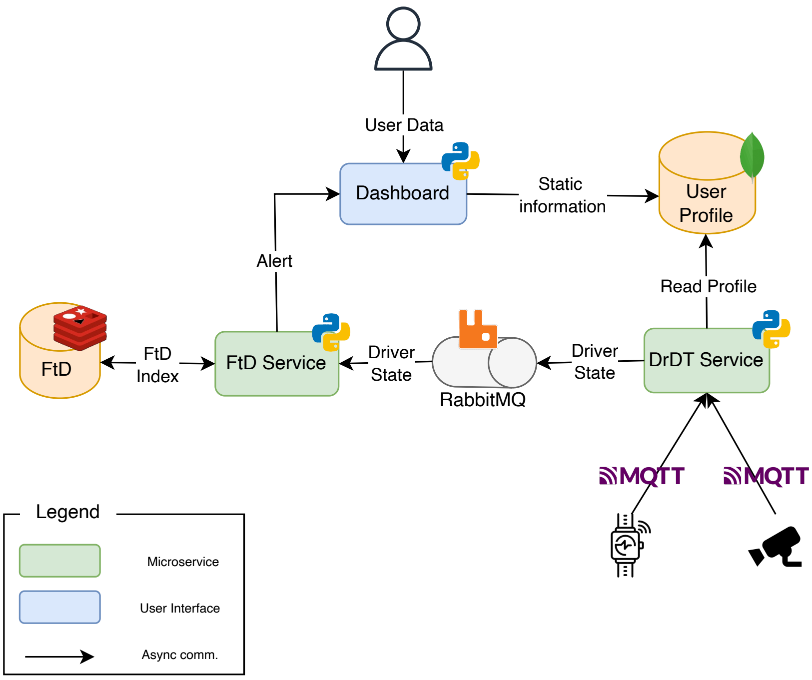

Figure 2. DrDT and Fitness-to-Drive microservice architecture.

3.3.1. Distraction Factor

The distraction factor DF captures both visual and cognitive distraction through an exponential saturation model:

$$DF = \sum_{i \in \{v,c\}} w_i \cdot D_i \cdot \left(1 - e^{-\lambda_i \cdot T_i}\right) \tag{2}$$

where D_i is the detected distraction level for type i (visual v or cognitive c), T_i is its duration, λ_i is a decay parameter, and $w_c + w_v = 1$. The exponential term models a critical behavioral insight: the *marginal* impact of distraction on fitness diminishes as the episode lengthens. This produces sharp early penalties — triggering timely alerts — while avoiding unbounded accumulation over prolonged episodes.

3.3.2. Emotional Factor

The emotional factor EF weights each of the seven emotions identified by Ekman [16] by their arousal level and driving relevance:

$$EF = \sum_{j \in \text{Emotions}} w_{E_j} \cdot A_j \cdot E_j \tag{3}$$

where E_j is the detected intensity of emotion j, A_j is its arousal level, and w_{E_j} is an empirically derived weight reflecting the emotion's impact on driving performance [43]. High-arousal negative emotions such as anger and fear carry larger weights than low-impact states such as happiness, capturing their disproportionate effect on reaction time and decision-making.

3.4. Implementation Details

The DrDT and FtD pipeline is realized as an event-driven microservice architecture (Figure 2), structured across three cyber-physical layers: a *physical layer* comprising the driver, the vehicle, and the IoT sensors; a *communication layer* handling real-time data transport between physical and cyber components; and a *cyber layer* responsible for twin state management, FtD computation, and adaptive feedback. The architecture is designed around three principles: low-latency data ingestion from heterogeneous sensors, clean separation between business logic and infrastructure, and hybrid storage optimized for both real-time access and historical analytics.

Messaging Infrastructure.

Sensor data arrives at two different timescales, and the messaging layer is designed accordingly. High-frequency physiological and behavioral streams, heart rate from smartwatch and emotion/distraction events from camera, are handled via MQTT, a lightweight publish-subscribe protocol well-suited to low-latency, continuous data flows. Each sensor type is served by a dedicated microservice adapter that normalizes raw readings before publishing them to the appropriate MQTT topic.

Inter-service communication between the DrDT service and the FtD service is handled by RabbitMQ [44], which uses the Advanced Message Queuing Protocol (AMQP) to guarantee reliable, ordered delivery of Driver State events. This separation, i.e., MQTT for sensor ingestion, RabbitMQ for coordination, ensures that high-frequency sensor noise does not interfere with the integrity of downstream FtD computation.

Processing Core and API Layer.

Each microservice is implemented in Python following a hexagonal architecture, which enforces a strict boundary between domain logic and external dependencies such as sensors, message brokers, and storage backends. This boundary makes it straightforward to swap or extend individual components, e.g., replacing the current smartwatch with a different physiological sensor, without touching the core FtD computation logic. An asynchronous event loop handles concurrent data streams from multiple sources without blocking, which is critical given the heterogeneous update rates of the inputs.

External access to the digital twin's state is exposed through a FastAPI-based RESTful API layer. This allows third-party applications, dashboards, and vehicle systems to query or update the DrDT in real time, enabling the broader integration of the framework into intelligent transportation ecosystems.

Storage.

Data persistence is handled through a two-tier hybrid strategy. MongoDB [45] serves as the primary store for sensor logs, driver profile records, and historical analytics, chosen for its flexible schema support — important given the evolving structure of driver profile data. Redis [46] complements this as an in-memory cache for frequently accessed values such as the most recently computed FtD index, ensuring sub-millisecond retrieval where latency is critical. Together, the two tiers cover the full temporal spectrum of the system's data needs: Redis for the present, MongoDB for the past.

4. Experimental Evaluation

We validate the proposed DrDT framework through controlled in-lab driving tests designed to assess whether personalization via Driver Profile genuinely improves FtD prediction accuracy over a generic baseline. We describe the experimental setup, ground-truth derivation, and results in turn.

4.1. Experimental Setup

All sessions were conducted using the CARLA driving simulator [47], running on the Town10 map[2] with 5 vehicles and 10 pedestrians, using a BMW as participants' vehicle. CARLA published vehicle speed, collision events, and lane departure events via MQTT to the processing pipeline. Modules were orchestrated via Docker Compose, with per-subject parameter configuration (e.g., individual heart rate baselines). Inter-module communication was handled by an Eclipse Mosquitto MQTT broker. Figure 3 shows the physical simulator setup.

[2] https://carla.readthedocs.io/en/latest/

Figure 3. Simulator setup for laboratory tests.

Data Acquisition Modules.

Three dedicated modules provided real-time driver state monitoring:

- **Arousal:** Heart rate was collected via the *Fitbit Sense* smartwatch, forwarded over WebSocket to an MQTT bridge, and converted to an arousal value normalized against each participant's 60-second resting baseline.
- **Emotion:** Facial expressions were classified in real time using a CNN-based model [48], processing frames from a dedicated USB camera and publishing the recognized emotion class via MQTT.
- **Distraction:** Visual distraction was detected using a custom CNN (TensorFlow) trained on the Vicomtech DMD dataset [49], reporting distraction events from the same in-cabin camera feed via MQTT.

Participants and Procedure.

Twenty licensed drivers (ages 20–50) completed individual 25-minute sessions structured as follows:

1. Resting heart rate measurement (60 s) and container configuration.
2. Simulator familiarization.
3. Standardized driving tasks: following a moving vehicle; simulating a phone call (left and right hand); sending text messages (left and right hand); drinking from a bottle (left and right hand).

These tasks were designed to systematically induce visual and cognitive distraction at varying intensities. Participants showed measurable increases in driving errors, particularly collisions and lane departures, during texting and mirror-checking tasks, consistent with the distraction model in Equation 2.

4.2. Ground-Truth FtD Derivation

A key methodological requirement is a ground-truth FtD against which model predictions can be compared. We derive this directly from observed driving errors, without relying on subjective ratings or external assessors.

For each driver i completing n_i trials, we recorded a binary error outcome $y_{i,j} \in \{0, 1\}$ per trial j, and computed the observed error rate:

$$\hat{p}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{i,j} \tag{4}$$

Since our model is calibrated under the linear link $P(\text{error}) = 1 - \text{FtD}$, the ground-truth FtD follows directly by inversion:

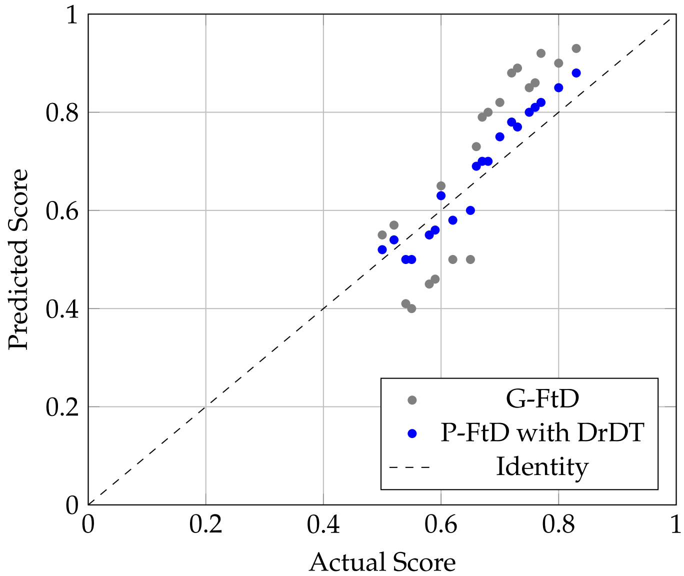

Figure 4. Ground-truth FtD vs. model-predicted FtD for the generic and personalized models. The dashed line $y = x$ denotes perfect agreement. Personalized predictions cluster markedly closer to the identity.

$$\mathrm{FtD}_i^{\mathrm{actual}} = 1 - \hat{p}_i \tag{5}$$

Since FtD varies continuously within each session, we summarize it as a temporal mean:

$$\mathrm{FtD}_i^{\mathrm{actual}} = \frac{1}{N_i} \sum_{j=1}^{N_i} \mathrm{FtD}_i(t_j) \tag{6}$$

where N_i is the number of equally-spaced time samples for driver i. This scalar ground truth is robust to transient fluctuations while remaining sensitive to systematic differences across participants. Prediction error for each model is then defined as:

$$\varepsilon_i = \mathrm{FtD}_i^{\mathrm{model}} - \mathrm{FtD}_i^{\mathrm{actual}} \tag{7}$$

from which we derive MAE and RMSE as our primary evaluation metrics.

4.3. Results

Figure 4 plots ground-truth FtD against model predictions for both the generic (G-FtD) and personalized (P-FtD) models. Points lying on the dashed identity line $y = x$ represent perfect prediction. The difference is immediately apparent: generic model predictions are broadly scattered around the identity, while personalized model predictions cluster tightly along it.

Quantitatively, the generic model yields a MAE of 0.11 and RMSE of 0.12. The personalized DrDT model reduces these to MAE $= 0.04$ and RMSE $= 0.045$ — a reduction of over **60%** and **55%** respectively. This result directly confirms the central hypothesis of this work: integrating individual driver characteristics via the Driver Profile substantially improves FtD prediction accuracy.

Figure 5 shows the distribution of absolute prediction errors for both models. The generic model has a median absolute error of 0.12 (IQR: 0.10–0.15, max: 0.16). The personalized model reduces the median to 0.05 (IQR: 0.03–0.05, max: 0.06). Crucially, the narrower box and shorter whiskers of the personalized model indicate not just lower error on average, but *more consistent* predictions across all participants, a property essential for a safety-critical system that must perform reliably for every driver, not just on average.

Figure 6 reinforces this picture through cumulative distribution functions. At an error threshold of 0.05, only 15% of generic model predictions qualify, compared to 90% of personalized model

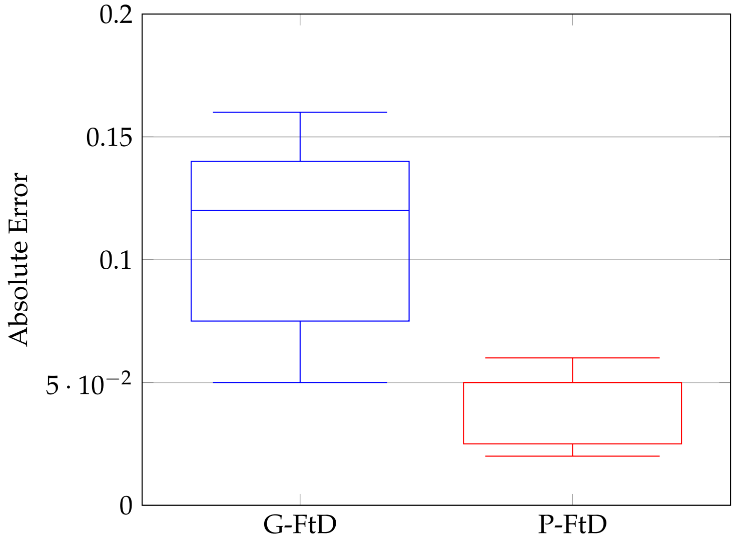

Figure 5. Absolute prediction error distributions for the generic (left) and personalized (right) models. Personalization reduces both the central tendency and spread of errors.

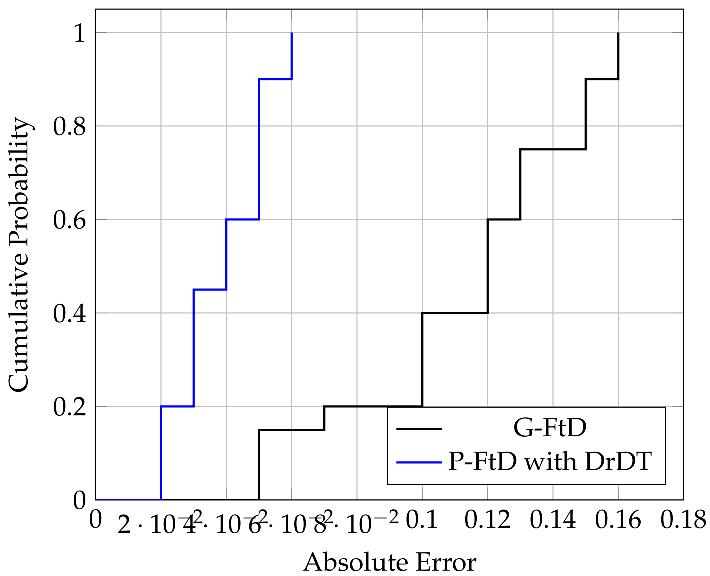

Figure 6. Cumulative distribution of absolute prediction errors. At a threshold of 0.05, the personalized model covers 90% of drivers versus 15% for the generic model.

predictions. To cover all 20 drivers, the generic model requires a tolerance of 0.16; the personalized model achieves the same with just 0.06. The personalized model's CDF is thus shifted dramatically leftward, confirming that its accuracy gains are consistent across the entire participant pool, not driven by a few well-predicted outliers.

Figure 7 reveals a theoretically important pattern: the accuracy gain from personalization grows monotonically with distance from the generic model's calibration mean (age 23.7 years). Drivers aged 20–30, closest to this mean, see the smallest improvement (ΔRMSE ≈ 0.03). Those aged 30–40 gain more (≈ 0.08), and the 40–50 group benefits most (≈ 0.11).

This finding is not merely a performance metric, i.e., it carries a direct safety implication. Generic models are by construction most accurate for the population they were calibrated on, and systematically less accurate for anyone who deviates from that profile. Since elderly drivers are among the highest-risk groups on the road [20], a monitoring system that performs worst precisely for those who need it most is fundamentally misaligned with road safety goals. The DrDT framework directly addresses

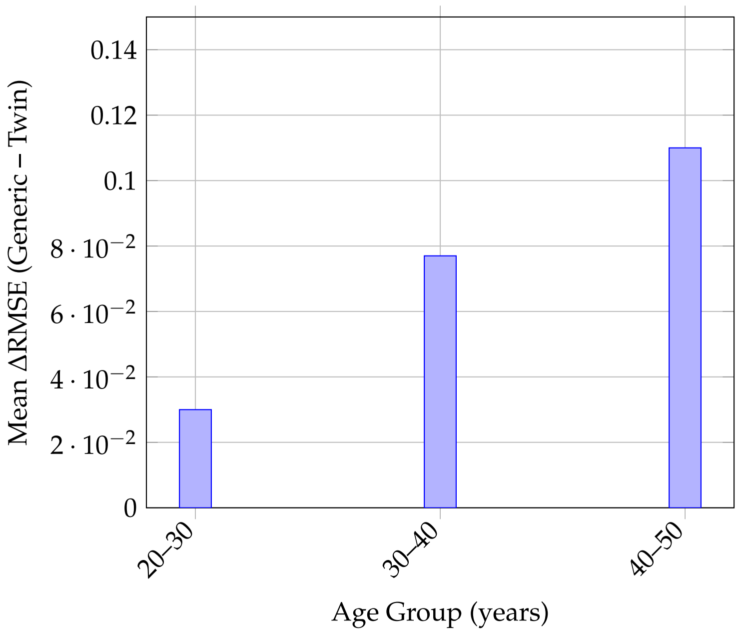

Figure 7. Mean RMSE improvement per age group. The benefit of personalization increases as driver age deviates from the generic model's calibration mean of 23.7 years.

this misalignment: by anchoring predictions to each driver's individual profile, it delivers the largest accuracy gains exactly where they matter most.

5. Discussion

The results presented in Section 4 demonstrate that personalization via the DrDT framework substantially improves FtD prediction accuracy. Here we reflect on what these findings mean for the design of adaptive cyber-physical driver monitoring systems, and on the limitations that must be addressed before real-world deployment.

5.1. Implications for Cyber-Physical Driver Monitoring

The core finding, i.e., personalization yields the largest gains precisely for drivers who deviate most from the population mean, has a direct design implication: a one-size-fits-all monitoring system is not merely suboptimal, it is systematically biased against the users who need it most. Elderly drivers, who face the highest per-violation accident risk [20], benefit most from the DrDT's individualization (ΔRMSE ≈ 0.11 for the 40–50 age group), yet are precisely the group most likely to be underrepresented in the calibration data of a generic model.

From a cyber-physical systems perspective, this finding has implications beyond prediction accuracy. In a CPS, the quality of the cyber representation directly determines the quality of the physical feedback loop: a twin calibrated to a population average will generate alerts that are systematically miscalibrated for any individual who deviates from that average, degrading the effectiveness of the physical intervention. The DrDT addresses this by anchoring the cyber state — and therefore the alert thresholds derived from it — to the individual driver's profile, ensuring that the physical-to-cyber-to-physical loop remains accurate across diverse drivers. This has particular relevance in the Internet of Vehicles context, where heterogeneous fleets of drivers with varying age, experience, and physiological baselines must be monitored by a shared infrastructure.

The dynamic profile management design, where the DrDT continuously refines its cyber representation of the driver over time, also opens the door to longitudinal adaptation at the IoT infrastructure level. A system that accumulates driver-specific data across sessions could detect gradual degradation in driving fitness associated with aging or health changes, trigger proactive interventions, and share anonymized behavioral patterns across a fleet to improve population-level models without sacrificing individual personalization. This positions the DrDT not just as a session-level monitor but as a persistent, continuously learning node in a broader intelligent transportation system.

From an IoT deployment standpoint, the event-driven microservice architecture demonstrated here (MQTT for sensor ingestion, RabbitMQ for service coordination, Redis for low-latency state access) provides a template that can scale horizontally across multiple vehicles and drivers. The hexagonal architecture ensures that sensor components can be swapped or extended as vehicle instrumentation evolves, without requiring changes to the core FtD computation logic. These properties are essential for real-world deployment in heterogeneous transportation environments.

5.2. Limitations

Several limitations of the current work must be acknowledged. First, the experimental sample comprises 20 participants aged 20–50, recruited in a controlled laboratory setting. While sufficient to demonstrate the proof-of-concept and establish statistical trends across age groups, this sample size limits the generalizability of the quantitative results. In particular, the 40–50 age group represents the oldest cohort in our study, yet the drivers most likely to benefit from personalization, e.g., those over 60, are entirely absent. Expanding the participant pool to include older drivers and a more diverse demographic range is a priority for future validation.

Second, all data were collected in a simulated environment using CARLA. Driving simulators, while valuable for controlled experimentation, differ from real-world driving in ways that may affect both the sensor signals and the behavioral responses of participants. Physiological arousal induced in a simulator may not fully replicate the stress profiles of on-road driving, and distraction tasks performed in a lab context may not capture the full complexity of naturalistic distraction. A real-world validation study, including assessment of IoT pipeline performance under realistic network conditions and sensor noise, is therefore an essential next step.

Third, the ground-truth FtD derivation relies on a linear mapping from observed error rate to the $[0, 1]$ FtD scale ($\mathrm{FtD}_i^{\mathrm{actual}} = 1 - \hat{p}_i$). While this provides a principled and objective reference, it is a simplification: driving fitness is a multidimensional construct, and binary error events capture only its most overt manifestation. Richer ground-truth signals — such as continuous physiological stress markers, near-miss detection, or vehicle dynamics anomalies — would provide a more comprehensive validation basis and are more naturally aligned with the continuous sensor streams available in an IoT-instrumented vehicle.

Finally, the current implementation does not yet model driving style (D_S) or driver confidence (C) as specified in Definition 1, and the Driver Profile is initialized from a fixed set of predefined archetypes rather than learned from data. Incorporating these dimensions would deepen the personalization of the cyber representation and is a priority for future instantiations of the framework. Additionally, real-world deployment will require addressing CPS-specific challenges including sensor calibration across heterogeneous vehicle platforms, latency guarantees under variable network conditions, and robust security mechanisms — including end-to-end encryption of physiological data streams and fine-grained access control — to meet the requirements of GDPR and related data protection frameworks.

6. Conclusion

We presented a Driver Digital Twin (DrDT) framework for computing a personalized Fitness-to-Drive (FtD) index that continuously adapts to each driver's physiological state, behavioral patterns, and individual profile. Built as a domain-specific instantiation of the Human Digital Twin, the DrDT integrates real-time signals, e.g. heart rate, facial emotion recognition, and visual distraction, with static and semi-static profile attributes including age, experience, and annual mileage, implemented as a low-latency microservice pipeline.

Experimental validation with 20 drivers in the CARLA simulator demonstrated that personalization yields substantial and consistent accuracy gains over a generic baseline: MAE reduced by over 60% (from 0.11 to 0.04) and RMSE by over 55% (from 0.12 to 0.05). Critically, these gains are not uniform as they grow monotonically with deviation from the generic model's calibration mean, reaching ΔRMSE ≈ 0.11 for the 40–50 age group. This confirms that personalization matters most for the drivers who are most poorly served by generic approaches, namely those who deviate from the population average, often the same groups at highest accident risk.

These results open several directions for future work. On the modeling side, the current DrDT does not yet instantiate driving style (D_S) or driver confidence (C) as defined in Definition 1; incorporating these dimensions could further sharpen FtD estimates, particularly in semi-autonomous driving contexts. On the learning side, the current Driver Profile relies on a fixed set of predefined archetypes for cold-start initialization. We plan to replace this with continual learning mechanisms that incrementally discover and refine archetypes from real-world data, enabling the system to adapt to behavioral patterns beyond those anticipated at design time. On the deployment side, real-world adoption will require addressing sensor calibration across vehicle platforms, network reliability under connectivity constraints, and user acceptance — particularly regarding the continuous collection of physiological and behavioral data. Privacy and security are non-negotiable requirements in this context, demanding robust encryption, fine-grained access controls, and full compliance with data protection regulations such as GDPR.

Ultimately, the core contribution of this work extends beyond the specific FtD metric: it establishes the DrDT as a principled, extensible framework for personalized driver monitoring. As vehicles become increasingly instrumented and automated, the ability to maintain an accurate, continuously updated model of the individual driver rather than an average one, will be central to building safety systems that work for everyone.

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